

T0-Model: Complete Parameter-Free Particle Mass Calculation Direct Geometric Method vs. Extended Yukawa Method With Complete Neutrino Quantum Number Analysis and QFT Derivation

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- Electron, Yukawa method: The calculation used the value $r_e \approx 1.349$ back-calculated from m_e instead of the fixed quantum number $4/3$. With $r_e = 4/3$: $m_e = 0.505$ MeV (-1.18% , accuracy 98.82% instead of 99.98%).
- Direct method (electron, muon): It is exactly equivalent to the Yukawa method via $f_i = 1/(r_i \xi^{p_i+1} \nu)$ (Eq. 52). In the worked example, however, f_i was set equal to r_i ($f_e = 1$, $f_\mu = 16/5$); this gave $1/\xi_e = 7500$ and $1/\xi_\mu = 2343.75$, with the measured values written next to them. Correct: $f_e = 1.485 \times 10^7$, $f_\mu = 7.146 \times 10^4$ GeV $^{-1}$.
- Quarks: strange $26/9 \cdot \xi \cdot \nu = 94.76$ MeV (-0.26%) instead of 95.0 MeV (0.00%); charm $2\xi^{2/3}\nu = 1.284$ GeV ($+0.32\%$) instead of 1.279 GeV (-0.08%). Average unchanged at 99.6% .
- Table “Numerical equivalence”: It used r_i back-calculated from measured values (1.349 etc.); now with the fixed quantum numbers and the f_i converted from them. Both methods give the same masses.
- 4th generation at 5.7 GeV: the arithmetic is right, but such a charged lepton is excluded by LEP.
- Established lepton patterns: $r_e \approx 1.35 \rightarrow 4/3$ (back-calculated value had remained).
- (1 Oct. 2026) Higgs check relation $\xi = \lambda_h^2 \nu^2 / (16\pi^3 m_h^2)$: with the SM self-coupling $\lambda_h = m_h^2 / (2\nu^2) \approx 0.129$ instead of the rounded 0.13 one obtains $\xi \approx 1.30 \times 10^{-4}$ (-2.4% from $4/30000$) instead of 1.318×10^{-4} (-1.15%); a consistency check of ξ , not an exact derivation (Doc. 385).
- (2 Oct. 2026) Notation f : structure factor $f(n, l, j)$ — not a frequency; note in the text (Doc. 190, R147).

The status of the neutrino part is given in the addendum at the end (R105, R109).

Abstract

The T0-Model offers two mathematically equivalent but conceptually different calculation methods for particle masses: the direct geometric method and the extended Yukawa method. Both approaches work with the single geometric constant $\xi = \frac{4}{3} \times 10^{-4}$; the rational quantum numbers (r_i, p_i) are structural choices, and ν serves as a conversion anchor. This documentation also contains neutrino quantum numbers and the quantum field theoretical derivation of the ξ -constant through EFT matching and 1-loop calculations. The neutrino part with double ξ -suppression is not tenable according to R109 (see the addendum at the end); for neutrinos, Doc. 340 is authoritative. For charged leptons and quarks, the average

deviation is below 1%; the over twenty free Standard Model mass parameters are thereby traced back to the single parameter ξ and fixed rational quantum numbers.

.1 Introduction

Particle physics faces a fundamental problem: The Standard Model with its over twenty free parameters offers no explanation for the observed particle masses. They are measured as inputs, not derived from a deeper principle. The T0-Model addresses this with two complementary calculation methods that work with the single parameter ξ and rational quantum numbers; the originally included treatment of neutrino masses is not tenable according to R109 (see addendum).

The Parameter Problem of the Standard Model

Despite its experimental success, the Standard Model has a theoretical gap: It contains more than 20 free parameters that must be determined experimentally. These include:

- **Fermion Masses:** 9 charged lepton and quark masses
- **Neutrino Masses:** 3 neutrino mass eigenvalues
- **Mixing Parameters:** 4 CKM and 4 PMNS matrix elements
- **Gauge Couplings:** 3 fundamental coupling constants
- **Higgs Parameters:** Vacuum expectation value and self-coupling
- **QCD Parameters:** Strong CP phase and others

Parameter Reduction

The T0-Model traces the over twenty free parameters of the Standard Model back to **one**: the geometric constant $\xi = \frac{4}{3} \times 10^{-4}$, motivated by the geometry of three-dimensional space. In addition there are the rational quantum numbers (r_i, p_i) as structural choices and ν as a conversion anchor. This version also contains neutrino quantum numbers (not tenable according to R109, see addendum) as well as the quantum field theoretical derivation.

.2 Methodological Clarification: Establishment vs. Prediction

Scientific-Historical Classification

The T0-Model draws on the methodology of **pattern recognition and systematic classification**, which was successful, for example, for the periodic table (Mendeleev 1869) or the quark model (Gell-Mann 1964).

Two-Phase Development

Phase 1: Establishment of Systematics

1. Pattern recognition in known particle masses (electron, muon, tau)
2. Parameter determination from experimental data
3. Establish quantum number assignment

4. Demonstrate mathematical equivalence of both methods

Phase 2: Unfolding Predictive Power

1. Extrapolation to unknown particles
2. Derive quark sector from lepton patterns
3. Predict new generations
4. Conduct experimental tests

Historical Precedence of Successful Pattern Physics

The comparison concerns the method of pattern recognition, not the state of confirmation:

Discovery	Pattern Recognition	Predictions	Confirmation
Periodic Table (1869)	Atomic weights and properties	Gallium, Germanium, Scandium	Experimentally confirmed
Spectral Lines (1885)	Hydrogen lines	Rydberg formula for all series	Quantum mechanics
Quark Model (1964)	Hadron masses	Eightfold Way	QCD theory
T0-Model (2025)	Lepton masses	4th generation, quarks	Experimental tests

Table 1: Historical precedence of pattern physics

.3 From Energy Fields to Particle Masses

The Fundamental Challenge

A central result of the T0-Model is that particle masses can be calculated from geometric principles. While the Standard Model requires over 20 free parameters to describe particle masses, the T0-Model works with the geometric constant $\xi_{\text{geom}} = \frac{4}{3} \times 10^{-4}$ and rational quantum numbers; the deviations for charged leptons and quarks are below 1.2%.

Parameter Reduction for Masses

Parameter Reduction:

- **Standard Model:** 20+ free mass parameters (measured)
- **T0-Model:** 1 parameter ξ plus rational quantum numbers (geometric)
- **Experimental Accuracy:** 99% average agreement (charged leptons and quarks; neutrino part see addendum)
- **Theoretical Foundation:** Three-dimensional spatial geometry + QFT derivation

Energy-Based Mass Concept

In the T0 framework, what we traditionally call "mass" is interpreted as an expression of characteristic energy scales of field excitations:

$$m_i \rightarrow E_{\text{char},i} \quad (\text{characteristic energy of particle type } i) \quad (1)$$

This reinterpretation treats mass and energy as different aspects of the same quantity.

.4 Two Complementary Calculation Methods

The T0-Model offers two mathematically equivalent but conceptually different approaches to calculating particle masses:

Method 1: Direct Geometric Resonance

Conceptual Foundation: Particles as resonances in the universal energy field

The direct method treats particles as characteristic resonance modes of the energy field $E(x, t)$, analogous to standing wave patterns:

$$\text{Particle} = \text{Discrete resonance modes of } E(x, t)(x, t) \quad (2)$$

Three-Step Calculation Process:

Step 1: Geometric Quantization

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i) \quad (3)$$

(Note of 2 Oct. 2026 on notation: here $f(n, l, j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

where:

$$\xi_0 = \frac{4}{3} \times 10^{-4} \quad (\text{geometric base parameter}) \quad (4)$$

$$n_i, l_i, j_i = \text{quantum numbers from 3D wave equation} \quad (5)$$

$$f(n_i, l_i, j_i) = \text{geometric function from spatial harmonics} \quad (6)$$

Step 2: Resonance Frequencies

$$\omega_i = \frac{c^2}{\xi_i \cdot r_{\text{char}}} \quad (7)$$

In natural units ($c = 1$):

$$\omega_i = \frac{1}{\xi_i} \quad (8)$$

Step 3: Mass Determination from Energy Conservation

$$E_{\text{char},i} = \hbar \omega_i = \frac{\hbar}{\xi_i} \quad (9)$$

In natural units ($\hbar = 1$):

$$E_{\text{char},i} = \frac{1}{\xi_i} \quad (10)$$

Method 2: Extended Yukawa Method

Conceptual Foundation: Bridge to Standard Model formulation

The extended Yukawa method maintains compatibility with Standard Model calculations while making Yukawa couplings geometrically determined rather than empirically fitted:

$$E_{\text{char},i} = y_i \cdot v \quad (11)$$

where $v = 246$ GeV is the Higgs vacuum expectation value (SM definition: $v \equiv (\sqrt{2} G_F)^{-1/2}$; no independent observable, Doc. 351/R112).

Geometric Yukawa Couplings:

$$y_i = r_i \cdot \left(\frac{4}{3} \times 10^{-4} \right)^{\pi_i} \quad (12)$$

Generation Hierarchy:

$$\text{1st Generation: } \pi_i = \frac{3}{2} \quad (\text{electron, up quark}) \quad (13)$$

$$\text{2nd Generation: } \pi_i = 1 \quad (\text{muon, charm quark}) \quad (14)$$

$$\text{3rd Generation: } \pi_i = \frac{2}{3} \quad (\text{tau, top quark}) \quad (15)$$

The coefficients r_i are simple rational numbers determined by the geometric structure of each particle type.

.5 Quantum Field Theoretical Derivation of the ξ -Constant

EFT Matching and Yukawa Coupling after EWSB

After electroweak symmetry breaking, we have the Yukawa interaction:

$$\mathcal{L}_{\text{Yukawa}} \supset -\lambda_h \bar{\psi} \psi H, \quad \text{with } H = \frac{v+h}{\sqrt{2}} \quad (16)$$

After EWSB:

$$\mathcal{L} \supset -m \bar{\psi} \psi - y h \bar{\psi} \psi \quad (17)$$

with the relations:

$$m = \frac{\lambda_h v}{\sqrt{2}} \quad \text{and} \quad y = \frac{\lambda_h}{\sqrt{2}} \quad (18)$$

The local mass dependence on the physical Higgs field $h(x)$ leads to:

$$m(h) = m \left(1 + \frac{h}{v} \right) \Rightarrow \partial_\mu m = \frac{m}{v} \partial_\mu h \quad (19)$$

T0 Operators in Effective Field Theory

In T0 theory, operators of the form appear:

$$O_T = \bar{\psi} \gamma^\mu \Gamma_\mu^{(T)} \psi \quad (20)$$

with the characteristic time field coupling term:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} \quad (21)$$

Substituting the Higgs dependence:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} = \frac{1}{mv} \partial_\mu h \quad (22)$$

This suggests that a $\partial_\mu h$ -coupled vector current is the UV origin.

1-Loop Matching Calculation

The 1-loop amplitude for the T0 vertex yields:

$$F_V(0) = \frac{y^2}{16\pi^2} \left[\frac{1}{2} - \frac{1}{2} \ln \left(\frac{m_h^2}{\mu^2} \right) + \frac{r(r - \ln r - 1)}{(r - 1)^2} \right] \quad (23)$$

For hierarchical masses ($m \ll m_h$), the constant term dominates:

$$F_V(0) \approx \frac{y^2}{32\pi^2} \quad (24)$$

§ Formula from Higgs Physics

The EFT matching yields the relationship:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \quad (25)$$

With Standard Higgs parameters ($m_h = 125.1$ GeV, $v = 246.22$ GeV, $\lambda_h = m_h^2/(2v^2) \approx 0.129$):

$$\xi \approx 1.30 \times 10^{-4} \quad (26)$$

This lies close to the geometric setting $\xi_0 = \frac{4}{3} \times 10^{-4} \approx 1.333 \times 10^{-4}$ (deviation $\approx -2.4\%$). The relation is a consistency check of ξ , not an exact derivation. *(Corrected on 1 Oct. 2026: previously $\lambda_h \approx 0.13$, $\xi \approx 1.318 \times 10^{-4}$, deviation $\approx 1.15\%$ – with the SM self-coupling $\lambda_h = m_h^2/(2v^2) = 0.1291$ one has $\xi = m_h^2/(64\pi^3 v^2) = 1.301 \times 10^{-4}$, -2.4% ; the rounded value 0.13 artificially lowered the deviation to about 1%; origin of the Higgs route in Doc. 385.)*

.6 Universal Particle Mass Systematics

Revised Universal Table of Fermions

Fermion	Generation	Family	Spin	r_f	Exponent p_f	SM Correspondence
Electron Neutrino	1	0	1/2	4/3	5/2	Double ξ
Electron	1	0	1/2	4/3	3/2	Lepton number
Muon Neutrino	2	1	1/2	16/5	3	Double ξ
Muon	2	1	1/2	16/5	1	Lepton number
Tau Neutrino	3	2	1/2	25/9	25/9	Double ξ
Tau	3	2	1/2	25/9	2/3	Lepton number
Up	1	0	1/2	6	3/2	Color
Down	1	0	1/2	$\frac{25}{2}$	3/2	Color + Isospin
Charm	2	1	1/2	2*	2/3	Color
Strange	2	1	1/2	$\frac{26}{9}$	1	Color
Top	3	2	1/2	$\frac{1}{28}$	-1/3	Color
Bottom	3	2	1/2	$\frac{3}{2}$	1/2	Color

.7 Numerical Reconstruction

The following analysis shows the explicit calculation of all fermions with both methods:

^{0*} Corrected from originally 8/9 based on detailed numerical analysis

Foundations and Experimental Input Data

Fundamental Constants:

$$\xi_0 = \xi = \frac{4}{3} \times 10^{-4} = 1.33333333 \dots \times 10^{-4} \quad (27)$$

$$\nu = 246 \text{ GeV} \quad (28)$$

Experimental Masses (PDG-near values):

$$m_e^{\text{exp}} = 0.0005109989461 \text{ GeV} \quad (29)$$

$$m_\mu^{\text{exp}} = 0.1056583745 \text{ GeV} \quad (30)$$

$$m_\tau^{\text{exp}} = 1.77686 \text{ GeV} \quad (31)$$

Charged Leptons: Detailed Calculations

Electron Mass Calculation:

Direct Method:

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2) \quad (32)$$

$$= \frac{4}{3} \times 10^{-4} \times 1 = \frac{4}{3} \times 10^{-4} \quad (33)$$

$$E_e = \frac{1}{\xi_e} = \frac{3}{4 \times 10^{-4}} = 7500 \quad (34)$$

Extended Yukawa Method:

$$r_e = \frac{4}{3} \quad (35)$$

$$y_e = \frac{4}{3} \times \left(\frac{4}{3} \times 10^{-4} \right)^{3/2} = 2.053 \times 10^{-6} \quad (36)$$

$$E_e = y_e \times 246 \text{ GeV} = 0.505 \text{ MeV} \quad (37)$$

Muon Mass Calculation:

Direct Method:

$$\xi_\mu = \frac{4}{3} \times 10^{-4} \times f_\mu(2, 1, 1/2) \quad (38)$$

$$= \frac{4}{3} \times 10^{-4} \times \frac{16}{5} = \frac{64}{15} \times 10^{-4} \quad (39)$$

$$E_\mu = \frac{1}{\xi_\mu} = 2343.75 \quad (40)$$

Extended Yukawa Method:

$$y_\mu = \frac{16}{5} \times \left(\frac{4}{3} \times 10^{-4} \right)^1 = 4.267 \times 10^{-4} \quad (41)$$

$$E_\mu = y_\mu \times 246 \text{ GeV} = 104.96 \text{ MeV} \quad (42)$$

Experiment: 105.66 MeV → Deviation $\approx 0.65\%$

Experiment electron: 0.511 MeV → Deviation -1.18%

(Corrected on 30 Sept. 2026: previously the Yukawa method gave $r_e = m_e^{\text{exp}} / (\nu \xi^{3/2}) \approx 1.349$ and $E_e = 0.511 \text{ MeV}$. That is the prefactor back-calculated from the measured value; the fixed quantum number of the tables is $r_e = 4/3$ and gives 0.505 MeV. The direct method gave $E_e = 1/\xi_e = 0.511 \text{ MeV}$ and $E_\mu = 1/\xi_\mu = 105.66 \text{ MeV}$ with $f_e = 1$ and $f_\mu = 16/5$. These are the Yukawa prefactors r_i , not the factors f_i of the direct method; they yield the dimensionless numbers 7500 and 2343.75. Via Eq. 52, $f_e = 1/(r_e \xi^{5/2} \nu) = 1.485 \times 10^7 \text{ GeV}^{-1}$ and $f_\mu = 1/(r_\mu \xi^2 \nu) = 7.146 \times 10^4 \text{ GeV}^{-1}$; with these the direct method gives $1/(\xi f_e) = 0.505 \text{ MeV}$ and $1/(\xi f_\mu) = 104.96 \text{ MeV}$, the same values as the Yukawa method.)

Neutrino Treatment

[Neutrino Approach] This section proposes a geometric treatment of neutrino masses via a **double ξ -suppression**. According to R109, this neutrino part is not tenable (see the addendum at the end); for neutrinos, Doc. 340 is authoritative.

Neutrino Quantum Numbers

Neutrinos follow the same quantum number structure as other fermions, but with a modification due to their weak interaction nature:

Neutrino	n	l	j	Suppression
ν_e	1	0	1/2	Double ξ
ν_μ	2	1	1/2	Double ξ
ν_τ	3	2	1/2	Double ξ

Table 2: Neutrino quantum numbers with characteristic double ξ -suppression

Double ξ -Suppression Mechanism

The approach assumes that neutrinos experience an additional geometric suppression factor:

$$f(n_{\nu_i}, l_{\nu_i}, j_{\nu_i}) = f(n_i, l_i, j_i)_{\text{Lepton}} \times \xi \quad (43)$$

Neutrino Mass Calculations:

Electron Neutrino:

$$\xi_{\nu_e} = \frac{4}{3} \times 10^{-4} \times 1 \times \frac{4}{3} \times 10^{-4} = \frac{16}{9} \times 10^{-8} \quad (44)$$

$$E_{\nu_e} = \frac{1}{\xi_{\nu_e}} = 9.1 \text{ meV} \quad (45)$$

Muon Neutrino:

$$\xi_{\nu_\mu} = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} \times \frac{4}{3} \times 10^{-4} = \frac{256}{45} \times 10^{-8} \quad (46)$$

$$E_{\nu_\mu} = \frac{1}{\xi_{\nu_\mu}} = 1.9 \text{ meV} \quad (47)$$

Tau Neutrino:

$$\xi_{\nu_\tau} = \frac{4}{3} \times 10^{-4} \times \frac{25}{9} \times \frac{4}{3} \times 10^{-4} = \frac{400}{81} \times 10^{-8} \quad (48)$$

$$E_{\nu_\tau} = \frac{1}{\xi_{\nu_\tau}} = 18.0 \text{ meV} \quad (49)$$

.8 Quark Analysis with Both Methods

Explicit Calculations of Quark Masses

We use $\xi = \frac{4}{3} \times 10^{-4}$ and $v = 246 \text{ GeV}$ (SM definition from G_F , Doc. 351/R112). For the Yukawa representation:

$$y_i = r_i \xi^{p_i}, \quad m_i^{\text{pred}} = y_i v.$$

For the direct geometric representation:

$$f_i = \frac{1}{\xi m_i^{\text{exp}}}, \quad m_i^{\text{exp}} = \frac{1}{\xi f_i}.$$

Quark	p_i	r_i (corr.)	m_i^{pred} (GeV)	m_i^{exp} (GeV)	rel. error (%)	Remark
Up	3/2	6	2.272×10^{-3}	2.27×10^{-3}	+0.11	OK
Down	3/2	25/2	4.734×10^{-3}	4.72×10^{-3}	+0.30	OK
Strange	1	26/9	9.476×10^{-2}	9.50×10^{-2}	-0.26	OK
Charm	2/3	2	1.284×10^0	1.28	+0.32	Corrected
Bottom	1/2	3/2	4.261×10^0	4.26	+0.02	OK
Top	-1/3	1/28	1.7198×10^2	171	+0.57	OK

Table 3: Yukawa predictions with corrected r_i , p_i and comparison with reference masses.

Correction for the Charm Quark

The originally specified value $r_c = 8/9$ does not reproduce the reference mass $m_c = 1.28$ GeV. The required value is:

$$r_c^{\text{required}} = \frac{m_c^{\text{exp}}}{v \xi^{2/3}} \approx 1.994 \approx 2.$$

Therefore, $r_c \approx 2$ was used in the corrected universal table. With $r_c = 2$ exactly, $m_c = 1.284$ GeV (+0.32 %).

(Corrected on 30 Sept. 2026: previously Table 3 gave strange 9.50×10^{-2} GeV (0.00 %, "Exact") and charm 1.279 GeV (-0.08 %). The formula written down with the table values gives $26/9 \cdot \xi \cdot 246 \text{ GeV} = 94.76 \text{ MeV}$ and $2 \xi^{2/3} \cdot 246 \text{ GeV} = 1.284 \text{ GeV}$; 1.279 GeV would correspond to $r_c = 1.994$.)

.9 Experimental Comparison

Accuracy Analysis

Comparison of the T0 values with the experimental masses:

Particle	T0 Prediction	Experiment	Accuracy	Type
<i>Charged Leptons</i>				
Electron	0.505 MeV	0.511 MeV	98.82%	Lepton
Muon	104.96 MeV	105.66 MeV	99.35%	Lepton
Tau	1783.4 MeV	1776.86 MeV	99.63%	Lepton
<i>Neutrinos</i>				
ν_e	9.1 meV	< 450 meV	Compatible	Neutrino
ν_μ	1.9 meV	< 180 keV	Compatible	Neutrino
ν_τ	18.0 meV	< 18 MeV	Compatible	Neutrino
<i>Quarks</i>				
Up quark	2.272 MeV	2.27 MeV	99.89%	Quark
Down quark	4.734 MeV	4.72 MeV	99.70%	Quark
Strange quark	94.76 MeV	95.0 MeV	99.74%	Quark
Charm quark	1.284 GeV	1.28 GeV	99.68%	Quark
Bottom quark	4.261 GeV	4.26 GeV	99.98%	Quark
Top quark	171.99 GeV	171 GeV	99.43%	Quark
Average			99.6%	All Fermions

Table 4: Comparison of T0 values with experimental masses

(Corrected on 30 Sept. 2026: previously electron 0.511 MeV (99.98%), strange 95.0 MeV (100.0%), charm 1.279 GeV (99.92%); recalculated with the fixed quantum numbers $r_e = 4/3$, $r_s = 26/9$, $r_c = 2$. The average remains 99.6%.)

Result for Charged Leptons and Quarks

The T0-Model achieves 99.6% average accuracy across the charged leptons and quarks, with the single parameter ξ and fixed rational quantum numbers. The neutrino rows are given only as compatibility with bounds; the neutrino part is not tenable according to R109 (see addendum).

.10 Predictive Power of the Established System

New Particle Generations

With the established patterns, new particles can be predicted:

4th Generation (extrapolated):

$$n = 4, \quad \pi_4 = \frac{1}{2}, \quad r_4 \approx 2.0 \quad (50)$$

$$m_{4\text{thGen}} = r_4 \times \xi^{1/2} \times v \approx 5.7 \text{ GeV} \quad (51)$$

(Note of 30 Sept. 2026: The arithmetic is right (5.68 GeV). A charged lepton of a 4th generation is excluded below about 100 GeV by LEP, and the Z width allows only three light neutrinos; this extrapolation is therefore not a viable prediction.)

Quark Sector Extrapolation

The lepton patterns can be transferred to quarks:

Quark	Generation	r_i	π_i	Prediction
Up	1	6	3/2	2.3 MeV
Down	1	12.5	3/2	4.7 MeV
Charm	2	2.0	2/3	1.3 GeV
Strange	2	2.89	1	95 MeV
Top	3	0.036	-1/3	173 GeV
Bottom	3	1.5	1/2	4.3 GeV

Table 5: Quark predictions from established patterns

.11 Corrected Interpretation of Mathematical Equivalence

Meaning of the Equivalence

The mathematical equivalence of both methods is **given by definition** when the parameters (r_i or f_i) are determined from the same experimental masses. The equivalence is not proof of the theory, but a consistency property of the mathematical structure.

Transformation Relationship as Bridge

The relationship:

$$f_i = \frac{1}{r_i \xi^{\pi_i} v \xi_0} \quad (52)$$

connects both methods mathematically. If r_i is determined from experimental masses, f_i follows automatically and vice versa.

Particle	m^{exp} (GeV)	r_i (Yukawa)	f_i (direct)	Accuracy
Electron	0.000511	4/3	1.485×10^7	98.82%
Muon	0.10566	16/5	7.146×10^4	99.34%
Tau	1.77686	25/9	4.205×10^3	99.63%
ν_e	9.1×10^{-6}	1.349	8.235×10^{10}	Prediction
ν_μ	1.9×10^{-6}	3.221	3.947×10^{11}	Prediction
ν_τ	18.0×10^{-6}	2.778	3.989×10^{10}	Prediction

Table 6: Numerical equivalence of both T0 methods for all leptons

(Corrected on 30 Sept. 2026: previously the charged-lepton rows had the $r_i = 1.349, 3.221, 2.778$ back-calculated from the measured values, with $f_i = 1.468 \times 10^7, 7.099 \times 10^4, 4.221 \times 10^3$ and accuracies 99.98, 99.35, 99.63 %. Now the fixed quantum numbers 4/3, 16/5, 25/9 are given with the f_i following from them via Eq. 52 (in GeV^{-1}); both methods give the same masses and hence the same accuracy. The neutrino rows are unchanged; for the neutrino part see addendum, R109.)

.12 Experimental Predictions and Precision Tests

Modified QED Vertex Corrections

T0 theory predicts modified Feynman rules:

$$\text{Time field vertex: } -i\gamma^\mu T_\mu^{(T)} = i\gamma^\mu \frac{\partial_\mu m}{m^2} \quad (53)$$

$$\text{Modified fermion propagator: } S_F^{(T0)}(p) = S_F(p) \cdot \left[1 + \frac{\beta}{p^2} \right] \quad (54)$$

Neutrino Validation

The T0 neutrino values are compatible with the experimental bounds listed here (on the tenability of the neutrino part, see addendum, R109):

Parameter	T0 Prediction	Experimental Limit	Status
m_{ν_e}	9.1 meV	< 450 meV (KATRIN)	Satisfied
m_{ν_μ}	1.9 meV	< 180 keV (indirect)	Satisfied
m_{ν_τ}	18.0 meV	< 18 MeV (indirect)	Satisfied
$\sum m_\nu$	29.8 meV	< 60 meV (Cosmology 2024)	Satisfied

Table 7: T0 neutrino predictions vs. experimental constraints

Neutrino Mass Hierarchy

The T0-Model predicts **normal ordering**: $m_{\nu_\mu} < m_{\nu_e} < m_{\nu_\tau}$, which is consistent with current oscillation data preferences.

.13 Methodological Foundation

Reversibility of the Established System

After the establishment phase, the T0 system can make predictions:

Established Lepton Patterns:

$$\text{1st Generation (n=1): } \pi_i = \frac{3}{2}, \quad r_e = \frac{4}{3} \approx 1.333 \quad (55)$$

$$\text{2nd Generation (n=2): } \pi_i = 1, \quad r_\mu = \frac{16}{5} = 3.2 \quad (56)$$

$$\text{3rd Generation (n=3): } \pi_i = \frac{2}{3}, \quad r_\tau = \frac{25}{9} \approx 2.778 \quad (57)$$

(Corrected on 30 Sept. 2026: previously $r_e \approx 1.35$ – this is the value back-calculated from the measurement, $m_e/(\nu\xi^{3/2}) = 1.349$; the quantum number is $r_e = 4/3$, as in the correction notes for the other places.)

Experimental Testability

The T0 predictions are experimentally falsifiable:

1. **LHC Searches:** New particles at characteristic energies (5-6 GeV range) (*note of 30 Sept. 2026: excluded by LEP for charged leptons, see section New Particle Generations*)
2. **Precision Measurements:** Refinement of r_i parameters
3. **Neutrino Tests:** Direct neutrino mass measurements
Methodologically, the following speaks for the T0 procedure:
 1. **Systematic Structure:** All parameters follow recognizable patterns
 2. **Predictive Power:** After establishment, new particles become predictable
 3. **Experimental Testability:** Predictions are falsifiable
 4. **QFT Foundation:** Quantum field theoretical derivation of the ξ -constant
 5. **Historical Models:** The methodology of pattern physics has been successful several times

.14 Parameter Economy and Universal Structure

No Adjustable Parameters

The scale of the T0 coefficients is determined by ξ , which can also be checked against Higgs parameters (deviation from $4/3 \times 10^{-4}$ about -2.4%):

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.30 \times 10^{-4} \quad (58)$$

(Corrected on 1 Oct. 2026: previously “can also be calculated”, about 1%, 1.318×10^{-4} – with $\lambda_h = m_h^2/(2v^2) \approx 0.129$ instead of 0.13 one obtains 1.30×10^{-4} ; consistency check, not a derivation, see Section .5.) Together with the rational quantum numbers (r_i, p_i) , no continuously adjustable parameter remains; v serves as a conversion anchor.

Universal Quantum Number Table

Particle	n	l	j	r_i	p_i	Special
<i>Charged Leptons</i>						
Electron	1	0	1/2	4/3	3/2	–
Muon	2	1	1/2	16/5	1	–
Tau	3	2	1/2	25/9	2/3	–
<i>Neutrinos</i>						
ν_e	1	0	1/2	4/3	5/2	Double ξ
ν_μ	2	1	1/2	16/5	3	Double ξ
ν_τ	3	2	1/2	25/9	25/9	Double ξ
<i>Quarks</i>						
Up	1	0	1/2	6	3/2	Color
Down	1	0	1/2	25/2	3/2	Color + Isospin
Charm	2	1	1/2	2	2/3	Color
Strange	2	1	1/2	26/9	1	Color
Top	3	2	1/2	1/28	-1/3	Color
Bottom	3	2	1/2	3/2	1/2	Color

Table 8: Universal quantum number table for all fermions

Winding numbers and topological assignment

The rational quantum numbers (r_i, p_i) are not arbitrary. They follow from the winding numbers (n_θ, n_ϕ) of the 4D torus T^4 via $r_i = n_\phi^2 / n_\theta$ (Doc. 317, August 2026):

Lepton	Gen.	n_θ	n_ϕ	r_i	p_i	KSAU knot
e	1	3	2	4/3	3/2	3_1 (trefoil)
μ	2	5	4	16/5	1	6_3 (amphicheiral)
τ	3	9	5	25/9	2/3	7_1 (torus knot)
ν_e	1	3	2	4/3	5/2	—
ν_μ	2	5	4	16/5	3	—
ν_τ	3	9	5	25/9	25/9	—

The exponent step $\Delta p = 1/3$ corresponds to the modulation frequency of the torus winding. The knot assignment (KSAU, Zenodo 2026) is **[S]**; the FFGFT quantum numbers are **[K]**. Details in Doc. 317 (KSAU-FFGFT synthesis).

.15 Critical Evaluation and Limitations

Theoretical Open Questions

1. **Number of Generations:** Why exactly three generations plus fourth prediction?
2. **Hierarchy Problem:** Connection between different energy scales
3. **CP Violation:** Incorporation of CKM and PMNS mixing matrices

.16 Concluding Assessment

Scientific Status

The T0-Model offers a systematic description of particle masses. The combination of:

- **High numerical accuracy** (99.6% across charged leptons and quarks)
- **Parameter economy** (one parameter ξ , rational quantum numbers)
- **Broad coverage** (charged leptons and quarks; neutrinos see addendum)
- **QFT consistency** (1-loop derivation of the ξ -constant)
- **Experimental testability** (specific falsifiable predictions)
makes the model a testable proposal.

Significance for Fundamental Physics

If experimentally confirmed, the T0-Model would considerably change our understanding of particle physics:

1. **Geometric Interpretation:** Particle masses as manifestations of 3D spatial geometry
2. **Unification:** All fermions follow the same universal structure
3. **Predictive Power:** New particles become predictable from established patterns
4. **Simplicity:** Strong simplification of complex phenomena

The T0-Model is an example showing that a unified description does not necessarily require greater complexity; the basic structure might be simple.

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Addendum: Status Clarification (7 September 2026)

R105: Retroactive marker certification (1 September 2026). This document was created before the marker system was introduced (Doc. A010) and carries no inline markers. Its derivations are internally consistent and serve as chain links for the Galois programme (Docs. 336, 338, 339).

Retroactive status for this document: Winding numbers r_i , exponents p_i , and the Yukawa method $m_i = r_i \xi^{p_i} v$ for charged leptons and quarks: **[K]**.

Dependent documents may treat these results as a secured chain link via reference to R105, without modifying this document.

R109: Restriction — neutrino part not viable [X] (1 September 2026). The “direct method” $E_i = 1/\xi_i$ for neutrinos is not a calculation: $1/(\xi r_i)$ gives the ratio $1 : 0.42$ for $e : \mu$ (inverse hierarchy, dimensionless); the MeV values alongside are measurements. The neutrino masses $9.1/1.9/18.0$ meV follow from three mutually incompatible prescriptions; the exponent $p_{\nu_\tau} = 25/9$ is r_{τ} , erroneously copied into the exponent column.

Valid: the Yukawa method $m_i = r_i \xi^{p_i} \nu$ for charged leptons and quarks ($< 1.2\%$ deviation). The **[K]** certification from R105 applies to this document *only* for r_i , p_i , and the Yukawa method — not for the neutrino part and not for the claimed equivalence of both methods.

Clarification (30 September 2026, R109): Both methods are exactly equivalent via $f_i = 1/(r_i \xi^{p_i+1} \nu)$ (Eq. 52) and give the same masses with fixed r_i , p_i . The only error was setting $f_i = r_i$ in the worked example; it has been corrected in the text.

Authoritative for neutrinos: Doc. 340 (R103, R108).